PRACTICAL- 10

AIM

To design a fuzzy inference system to estimate the house price based on house size and location quality using Python.

THEORY

Fuzzy Logic is a form of many-valued logic derived from fuzzy set theory that deals with reasoning that is approximate rather than fixed and exact. Unlike classical logic where values are either 0 (false) or 1 (true), fuzzy logic allows values between 0 and 1, representing degrees of truth.

Fuzzy logic is widely used in:

- Control systems
- Decision making
- Pattern recognition
- AI systems
- Recommendation systems

A **Fuzzy Inference System** maps inputs to outputs using fuzzy logic rules.

Main components:

1. **Fuzzification**
   - Converts crisp inputs into fuzzy values.
   - Example: House size = 1800 sq ft → Membership in *Medium* and *Large*.
2. **Rule Base**
   - Set of IF–THEN rules.
   - Example:
     IF size is Large AND location is Good → Price is High
3. **Inference Engine**
   - Applies fuzzy rules to determine output.
4. **Defuzzification**
   - Converts fuzzy output into a crisp numerical value.

SOFTWARE USED

Jupyter Notebook

CODE

```
import numpy as np

import skfuzzy as fuzz

from skfuzzy import control as ctrl


# Input variables

size = ctrl.Antecedent(np.arange(500, 4001, 1), 'size')

location = ctrl.Antecedent(np.arange(0, 11, 1), 'location')


# Output variable

price = ctrl.Consequent(np.arange(100000, 1000001, 1), 'price')


# Membership functions for house size

size['small'] = fuzz.trimf(size.universe, [500, 500, 1500])

size['medium'] = fuzz.trimf(size.universe, [1000, 2000, 3000])

size['large'] = fuzz.trimf(size.universe, [2500, 4000, 4000])


# Membership functions for location quality

location['poor'] = fuzz.trimf(location.universe, [0, 0, 5])

location['average'] = fuzz.trimf(location.universe, [3, 5, 7])

location['good'] = fuzz.trimf(location.universe, [6, 10, 10])


# Membership functions for price

price['low'] = fuzz.trimf(price.universe, [100000, 100000, 400000])

price['medium'] = fuzz.trimf(price.universe, [300000, 500000, 700000])

price['high'] = fuzz.trimf(price.universe, [600000, 1000000, 1000000])
```

```python
# Fuzzy rules
rule1 = ctrl.Rule(size['small'] & location['poor'], price['low'])
rule2 = ctrl.Rule(size['medium'] & location['average'], price['medium'])
rule3 = ctrl.Rule(size['large'] & location['good'], price['high'])
rule4 = ctrl.Rule(size['large'] & location['average'], price['high'])
rule5 = ctrl.Rule(size['medium'] & location['good'], price['high'])

# Control system
price_ctrl = ctrl.ControlSystem([rule1, rule2, rule3, rule4, rule5])
price_simulation = ctrl.ControlSystemSimulation(price_ctrl)

# Example input
price_simulation.input['size'] = 2000
price_simulation.input['location'] = 7

# Compute result
price_simulation.compute()

print("Estimated House Price:", price_simulation.output['price'])
```

OUTPUT

```
Estimated House Price: 823809.5238095256
```